

Tim N. Ortkamp

Doctoral Researcher in Applied Mathematics & Medical Physics

Location:	Heidelberg, Germany	Nationality:	German
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EDUCATION & RESEARCH

Visiting Researcher	2024 & 2026
<i>Department of Radiation Oncology, Massachusetts General Hospital</i>	
	<i>Boston, USA</i>

Doctoral Researcher	2021–2026
<i>Scientific Computing Center, Karlsruhe Institute of Technology & Department of Medical Physics in Radiation Oncology, German Cancer Research Center</i>	
	<i>Karlsruhe / Heidelberg</i>

Master of Science (M.Sc.) – Industrial Engineering	2018–2021
<i>Karlsruhe Institute of Technology</i>	
	<i>Karlsruhe</i>

Bachelor of Science (B.Sc.) – Industrial Engineering	2014–2018
<i>Karlsruhe Institute of Technology</i>	
	<i>Karlsruhe</i>

Abitur (University Entrance Qualification)	2006–2014
<i>Viktoriaschule Darmstadt</i>	
	<i>Darmstadt</i>

PROFESSIONAL EXPERIENCE

Student Assistant	2016–2021
<i>Institute for Operations Research, Karlsruhe Institute of Technology</i>	
	<i>Karlsruhe</i>

Intern in Data Science	2019
<i>Smart Data Analytics / MES & IIoT, Dürr Systems AG</i>	
	<i>Bietigheim-Bissingen</i>

Intern in Supply Chain Consulting	2015 & 2017
<i>Strategy and Concept Planning, Miebach Consulting GmbH</i>	
	<i>Frankfurt am Main</i>

PUBLICATIONS, PREPRINTS & OTHER WORKS

- [1] T. Ortkamp, H. Sallem, S. Harrabi, M. Frank, O. Jäkel, J. Bauer, N. Wahl. Direct optimization of the probability of lesion origin in proton treatment planning for low-grade glioma patients. *Medical Physics*, 2026. DOI: [10.1002/mp.70395](https://doi.org/10.1002/mp.70395)
- [2] T. Ortkamp, H. Sallem, S. Harrabi, M. Frank, O. Jäkel, J. Bauer, N. Wahl. Direct optimization of the probability of lesion origin in proton treatment planning for low-grade glioma patients. *arXiv preprint*, 2025. [arXiv:2506.13539](https://arxiv.org/abs/2506.13539) [physics.med-ph]
- [3] L. Voss, A. Neishabouri, T. Ortkamp, A. Mairani, N. Wahl. BayesDose: Comprehensive proton dose prediction with model uncertainty using Bayesian LST. *arXiv preprint*, 2023. [arXiv:2307.01151](https://arxiv.org/abs/2307.01151) [physics.med-ph]

Lecture Notes:

T. Ortkamp, M. Coblenz, O. Grothe. *Analyse multivariater Daten* (Analysis of Multivariate Data), 2021.

CONFERENCES & WORKSHOPS

- ESTRO Physics Workshop (Toulouse, 2025)
- ICCR 20 (Lyon, 2024, oral presentation & conference paper: [HAL: hal-04720234](https://hal.archives-ouvertes.fr/hal-04720234) (p. 841–844))
- EUROPT 21 (Lund, 2024, oral presentation)
- PTCOG 62 (Singapore, 2024, oral presentation)
- EUCCO 6 (Heidelberg, 2023, oral presentation)
- AAPM 65 (Houston, 2023, oral presentation)
- PTCOG 61 (Madrid, 2023, oral presentation)

OPEN-SOURCE SOFTWARE CONTRIBUTIONS

[pyanno4rt](#) (Lead Developer)

Python library for conventional and outcome prediction model-based inverse photon and proton treatment plan optimization.

[seamaze](#) (Lead Developer)

Python library for classical, limited-memory, and dynamical low-rank CMA-ES.

[pyRadPlan](#) (Contributor)

Radiotherapy treatment planning toolkit designed for interoperability with matRad.

TEACHING & ACADEMIC SUPERVISION

Co-supervisor

[10/2022–03/2023](#)

Seminar “Optimierung in der KI” (Optimization in AI), Scientific Computing Center, Karlsruhe Institute of Technology

Karlsruhe

Co-instructor & Assistant lecturer

[04/2022–08/2022](#)

Course “Optimierungstheorie” (Optimization Theory), Scientific Computing Center, Karlsruhe Institute of Technology

Karlsruhe

Instructor

[10/2019–03/2020](#)

Course “Analyse multivariater Daten” (Analysis of Multivariate Data), Institute for Operations Research, Karlsruhe Institute of Technology

Karlsruhe

Tutor

[10/2016–02/2017](#)

Course “Statistik II” (Statistics II), Institute for Operations Research, Karlsruhe Institute of Technology

Karlsruhe

Student Supervision

- *Proximale Algorithmen zur Optimierung von nicht-differenzierbaren Problemen in der Radiotherapie* (Proximal Algorithms for the Optimization of Non-Differentiable Problems in Radiotherapy) (Bachelor’s thesis, V. Deisner, 2025)
- *Machine Learning Model-Based Radiotherapy Treatment Plan Optimization Using Limited Memory Matrix Adaption Evolution Strategies* (Bachelor’s thesis, M. Müller, 2024)

- *Inverse Radiotherapy Treatment Planning using Machine Learning Outcome Prediction Models* (Student assistance, M. Müller, 2022–2024)
- *Machine Learning for Radiotherapy Outcome Prediction based on Imbalanced Data* (Bachelor's thesis, T. Thost, 2023)
- *Restringierte Optimierung von maschinellen Lernmodellen zur Vorhersage des Behandlungserfolgs in der Radiotherapie* (Constrained Optimization of Machine Learning Models for Outcome Prediction in Radiotherapy) (Bachelor's thesis, K. von Gaisberg, 2023)
- *Multikriterielle Optimierung und ihre Anwendung in der inversen Radiotherapieplanung* (Multi-Objective Optimization and its Application in Inverse Radiotherapy Treatment Planning) (Bachelor's thesis, L. Gseller, 2023)
- 2 co-supervisions at the German Cancer Research Center

SKILLS, AWARDS & INTERESTS

Languages

- *German*: Native
- *English*: Fluent
- *Swedish*: Basic

Awards

- Certificate for good teaching
Course “Analyse multivariater Daten” (Analysis of Multivariate Data), Karlsruhe Institute of Technology, 2020

Software & Programming

- *Languages*: Python, MATLAB, R
- *Tools & Markup*: Git, L^AT_EX, HTML/CSS

Personal Interests

- *Sports*: Cycling, Running, Soccer, Hiking
- *Music*: Piano, Synthesizer, Guitar, Songwriting, Audio recording/production